Adaptive Structured Pruning of Convolutional Neural Networks for Time Series Classification

Javidan Abdullayev, Maxime Devanne, Cyril Meyer, Ali Ismail-Fawaz, Jonathan Weber, Germain Forestier

IRIMAS, Université de Haute-Alsace

March 7, 2026

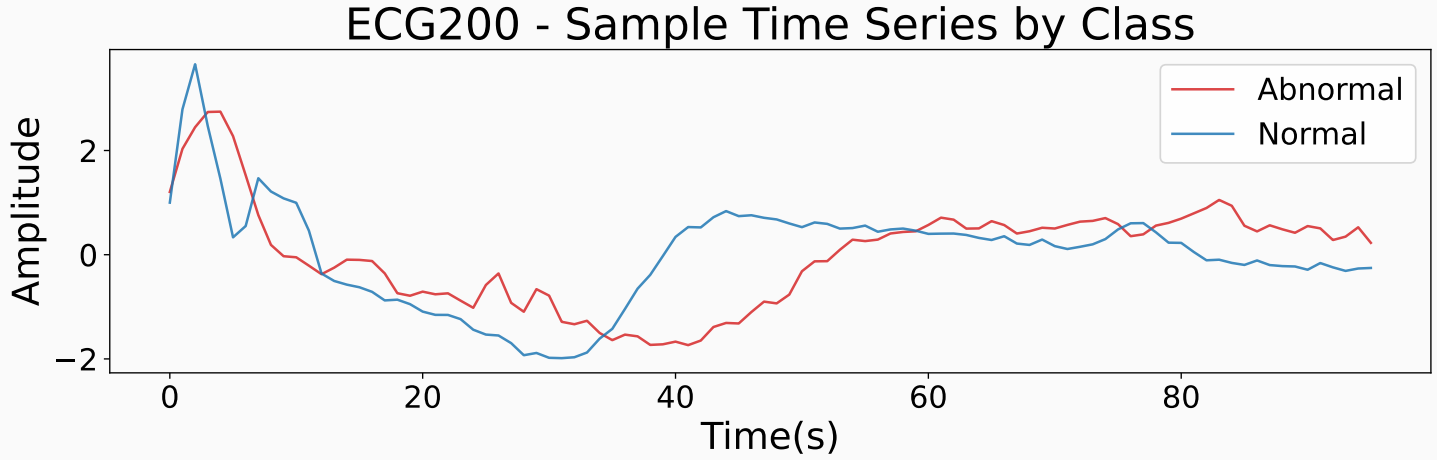
1. Introduction and Literature Review
2. Proposed Method
3. Experimental Evaluation

1 Introduction and Literature Review

2 Proposed Method

3 Experimental Evaluation

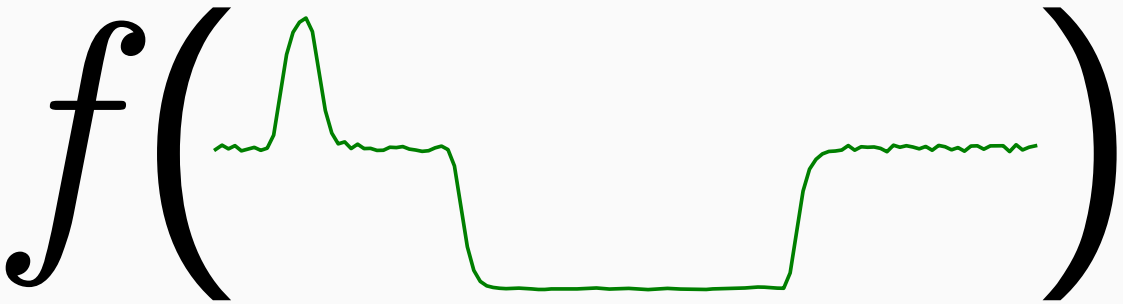
Time Series Data



Dau, Hoang Anh, et al. "The UCR time series archive." IEEE/CAA Journal of Automatica Sinica 6.6 (2019): 1293-1305.

Time Series Classification (TSC)

Time Series Classification Task



- f is a classifier
- f assigns a label y to an input sample
- $y \in \{0, \ldots, C-1\}$, with the C number of classes

Deep Learning Approaches for TSC

- Deep Learning (DL) for TSC

  Best performing DL approaches for TSC based on Convolutional Neural Networks

Middlehurst et al. "Bake off redux." Data Mining and Knowledge Discovery 38.4 (2024).

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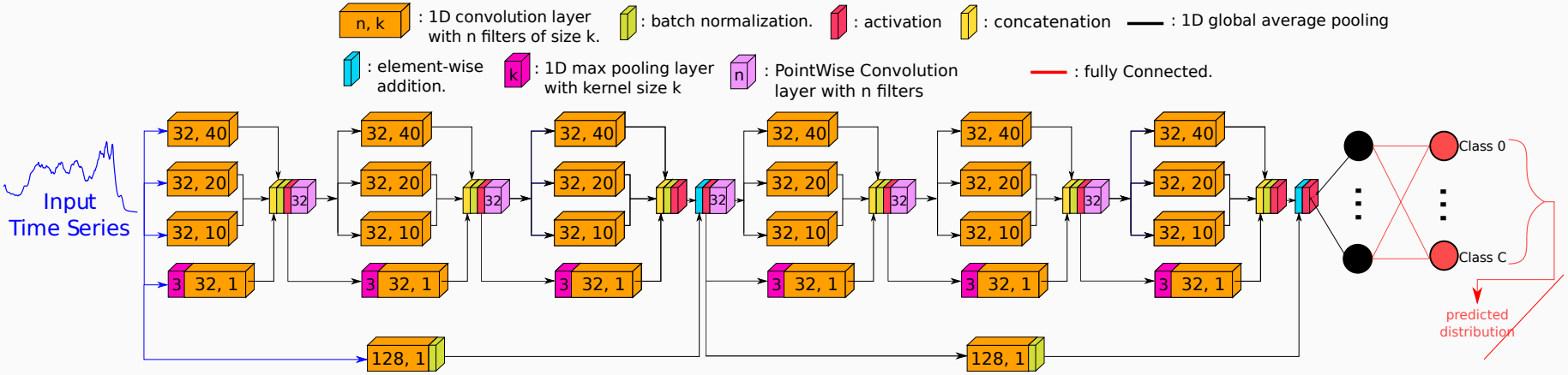

InceptionTime Architecture

Ismail Fawaz, Hassan, et al. "Inceptiontime: Finding alexnet for time series classification." Data mining and knowledge discovery 34.6 (2020): 1936-1962.

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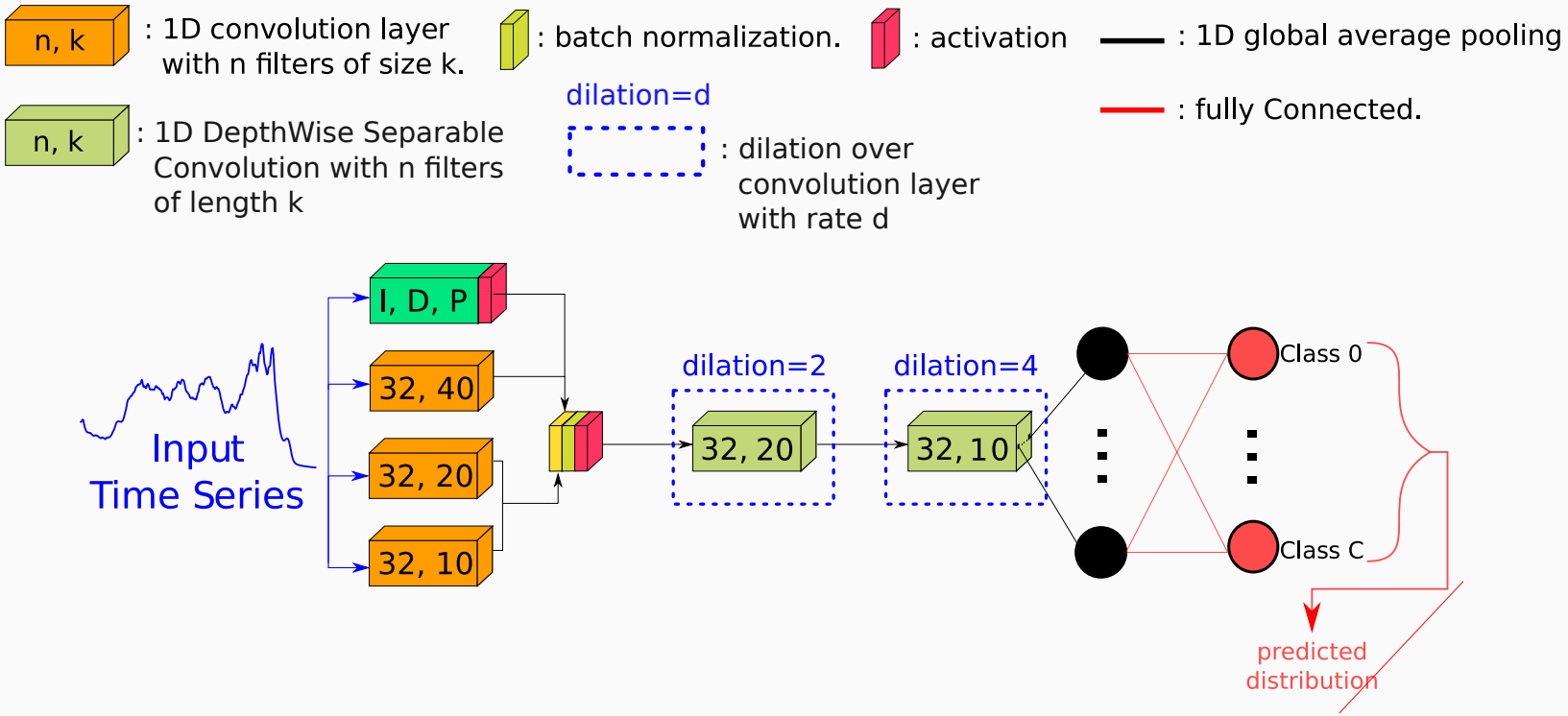

Lightweight InceptionTime Ensemble (LITE)

Ismail-Fawaz, Ali, et al. "Lite: Light inception with boosting techniques for time series classification." 2023 IEEE 10th International Conference on Data Science and Advanced Analytics (DSAA). IEEE, 2023.

Motivation

- TSC benchmarks show high dataset diversity:
  - 128 UCR datasets with varying complexity
  - Time series length: 24 – 2,844 | Classes: 2 – 60 | Samples: 16 – 8,926

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- **Research Question:**

  Can we automatically adapt model complexity to dataset difficulty without manual tuning of compression ratios?

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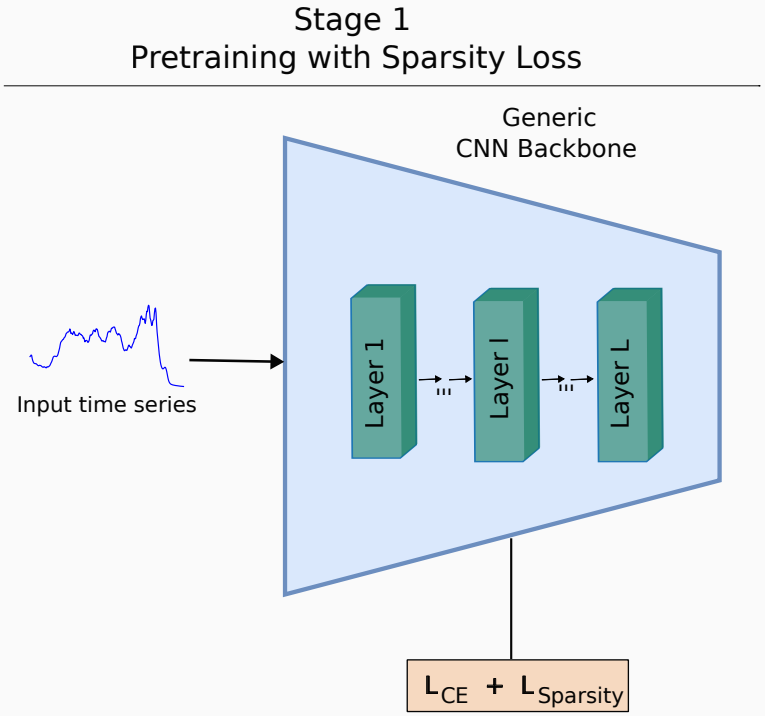
Proposed Approach

Overview of our Dynamic Structured Pruning Approach



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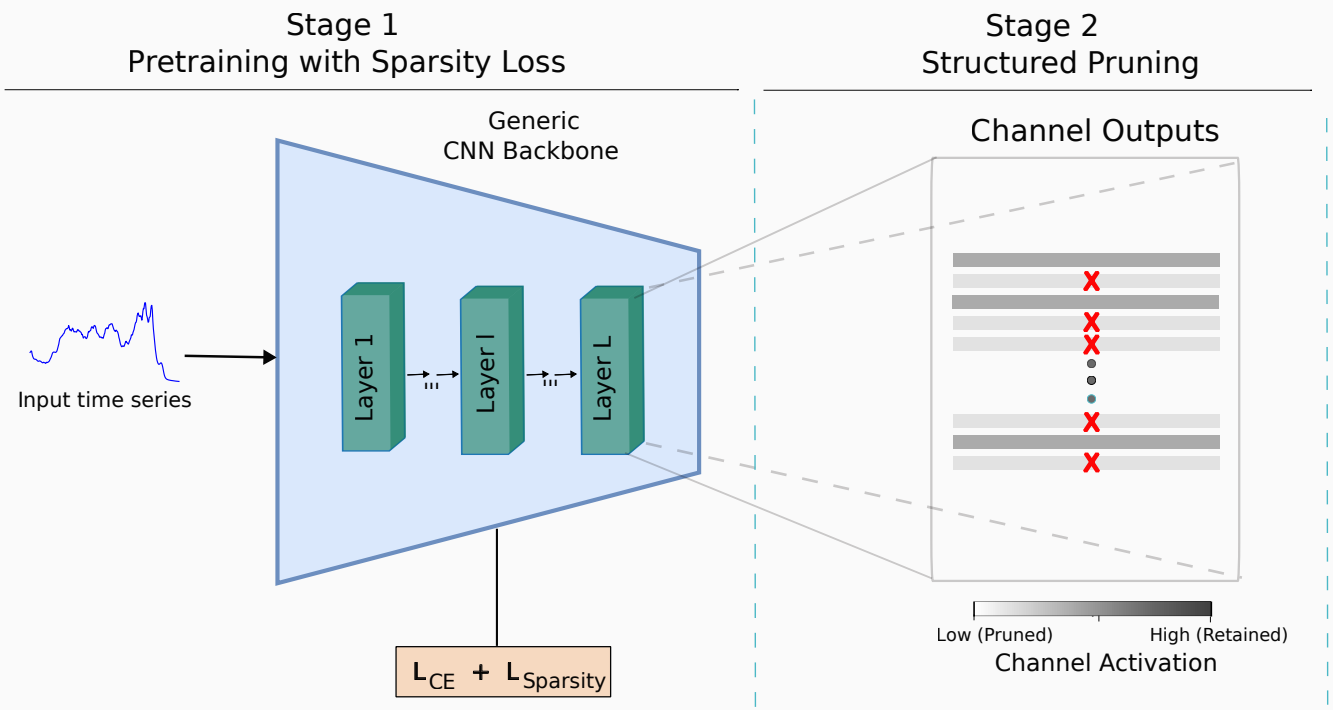

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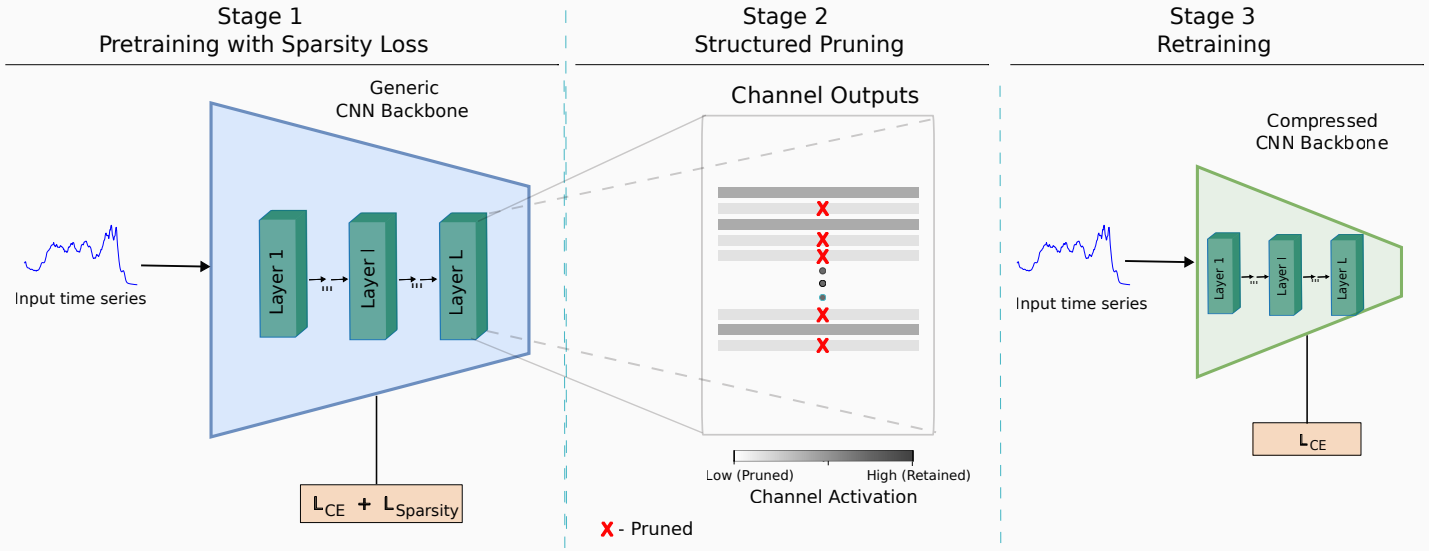
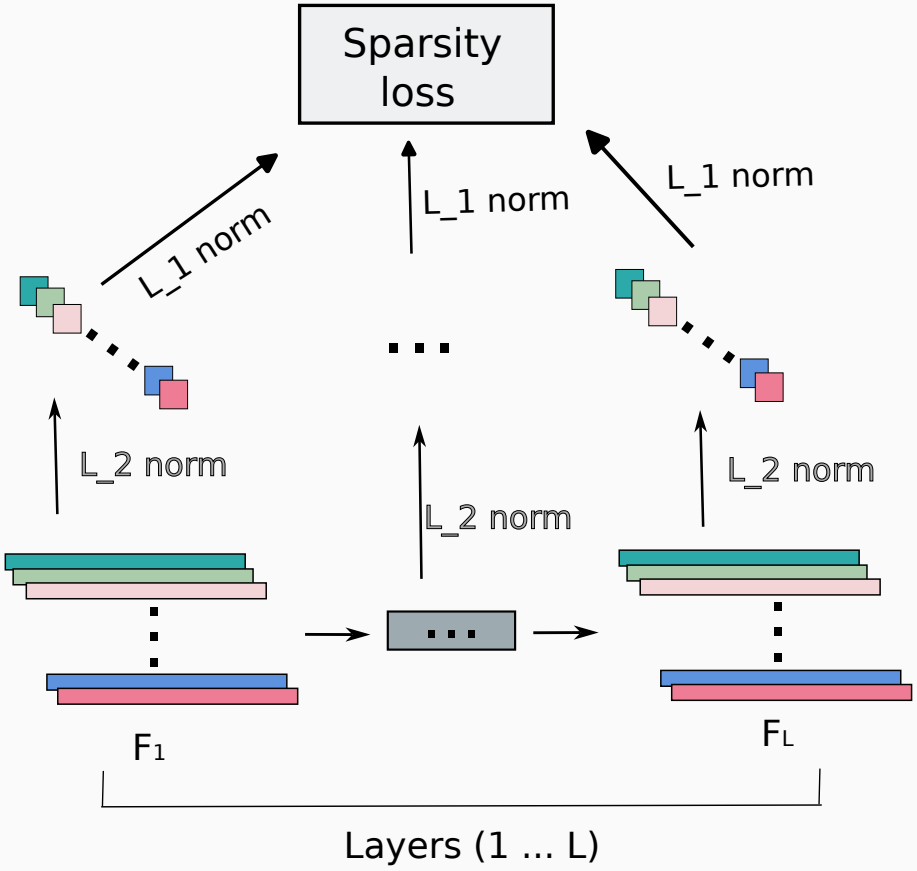


- ▷ **Stage 1:** Pretraining with *Auxiliary* Sparsity Loss - Promotes channel-level sparsity
- ▷ **Stage 2:** Structured Pruning - Prune low-activation filters globally
- ▷ **Stage 3:** Retraining - Maintains original accuracy

Stage 1: Pretraining with Sparsity Loss

Hierarchical Sparsity Loss:

$$\mathcal{L}_{\text{sparsity}} = \sum_{i=1}^{B} \sum_{j=1}^{C} \underbrace{\sqrt{\sum_{k=1}^{T} F[i,j,k]^2}}_{L_2 \text{ norm}}$$



Instance-wise sparsity loss induces channel-level sparsity

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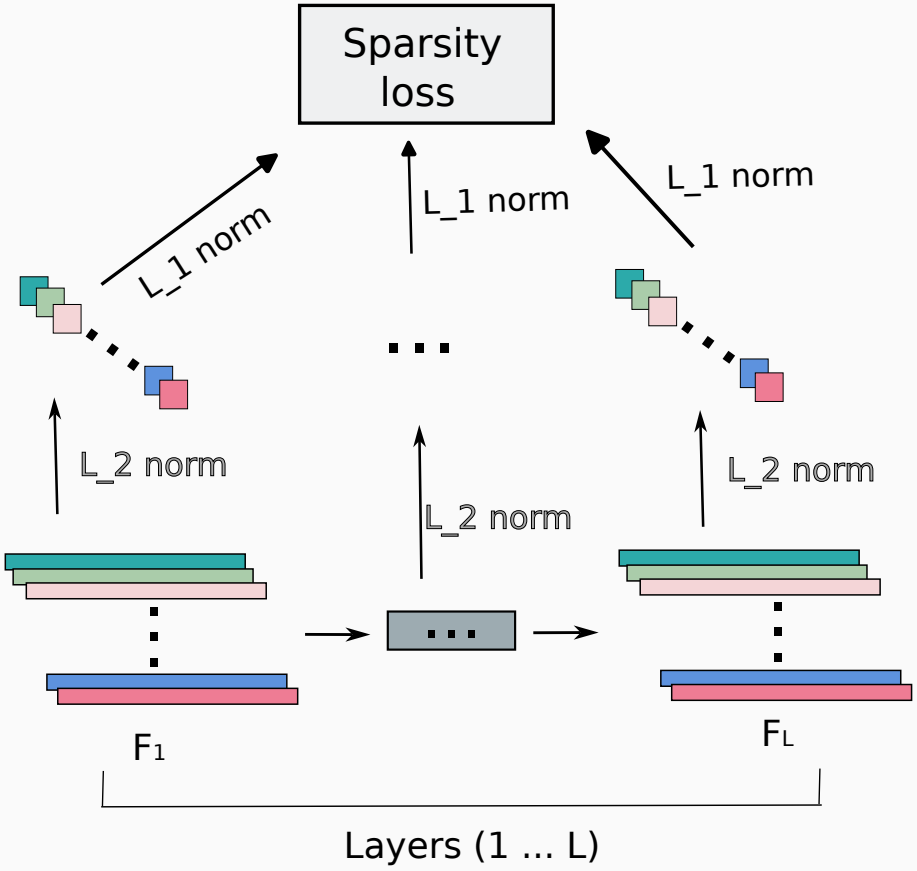
- L_2 norm: Measures channel energy
- L_1 norm: Enforces global sparsity
- Entire channels become inactive



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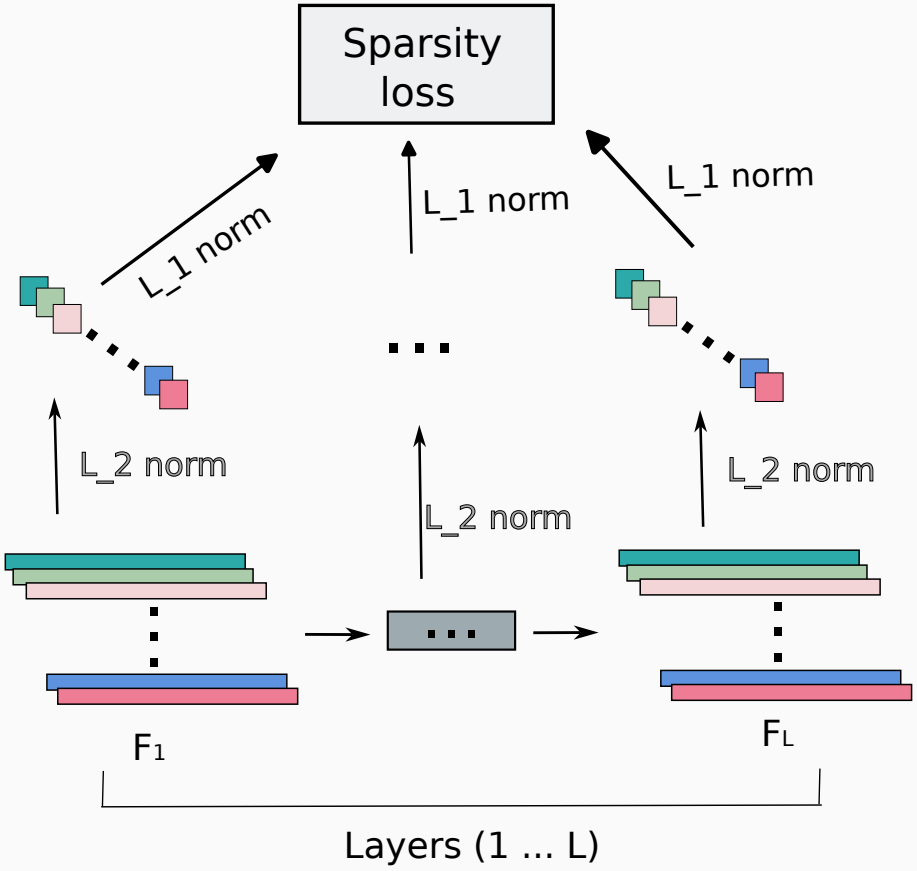
Training Objective:

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{CE} + \lambda \cdot \mathcal{L}_{\text{sparsity}}$$



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Stage 2: Global Activation Analysis & Pruning

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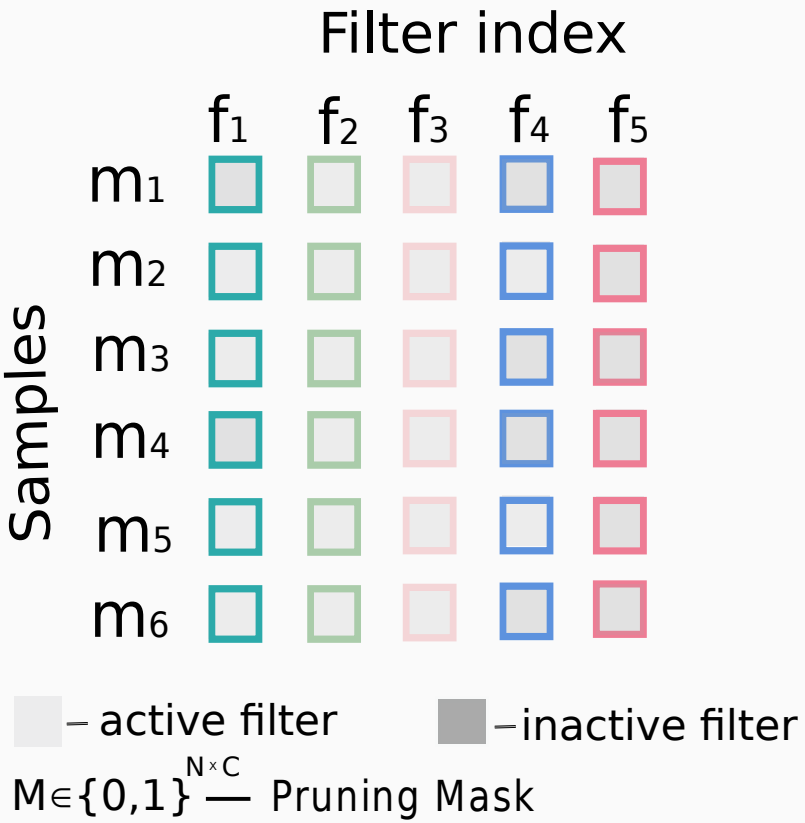

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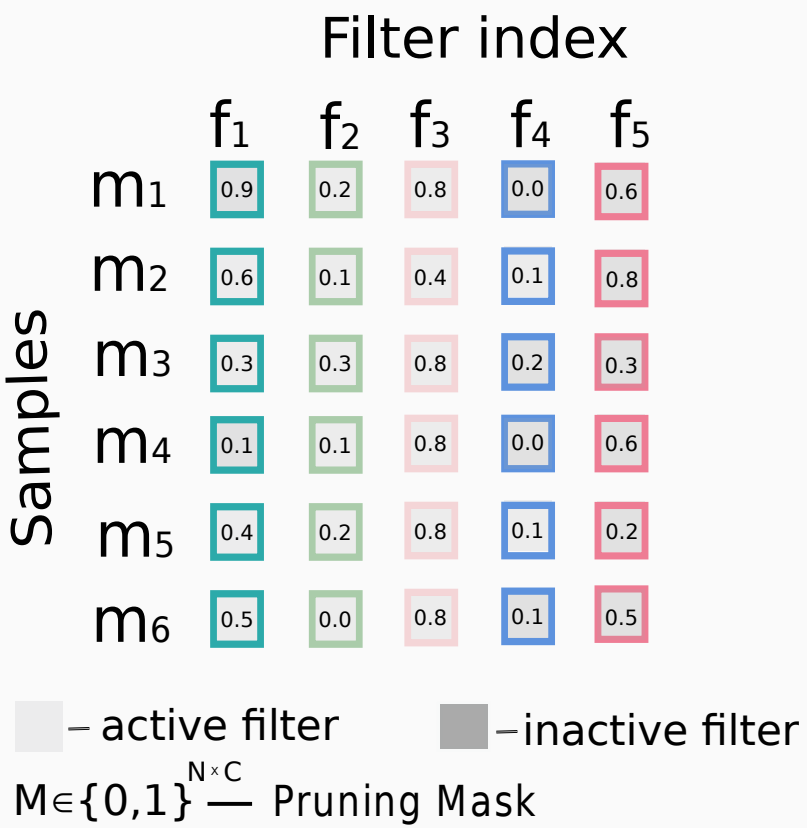

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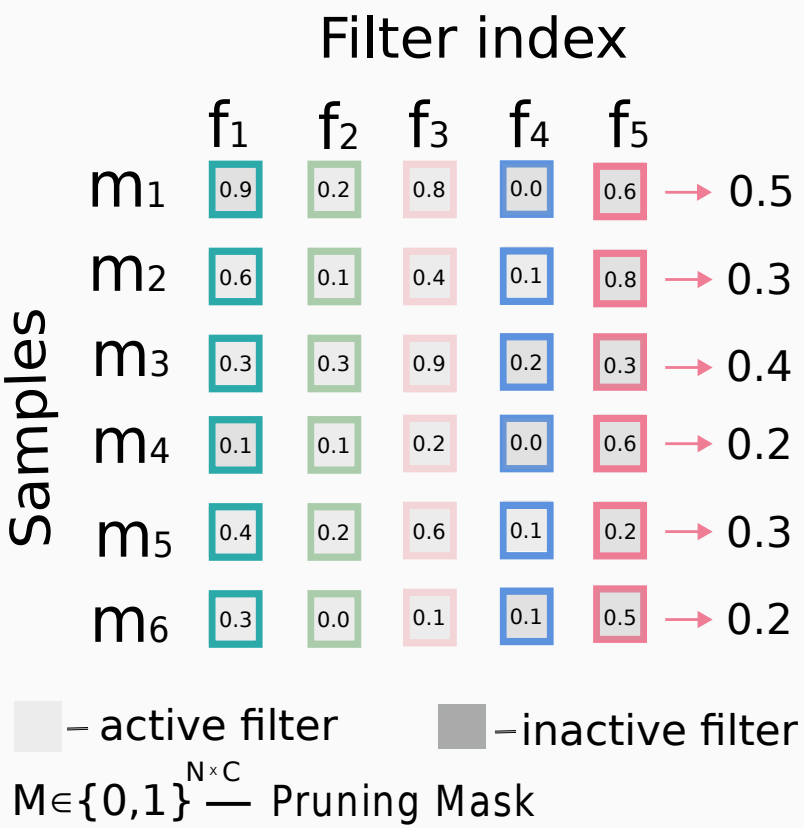

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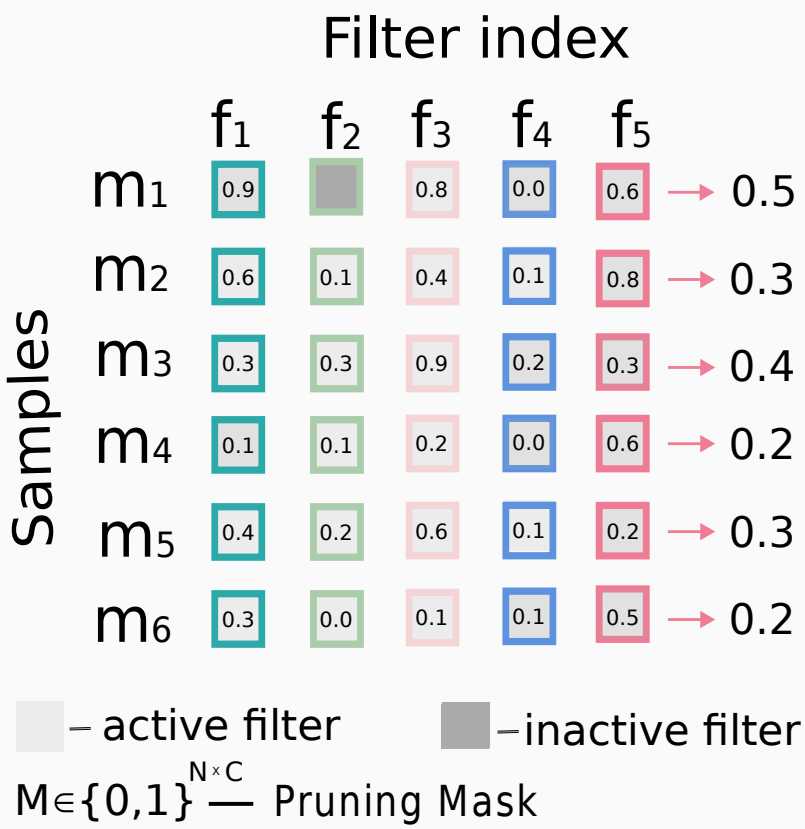

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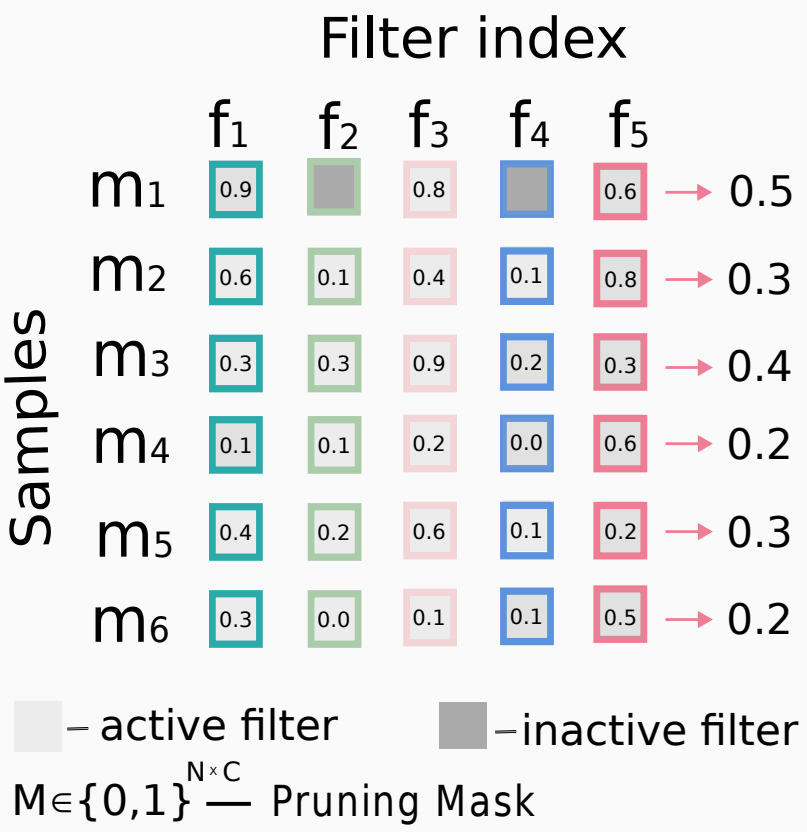

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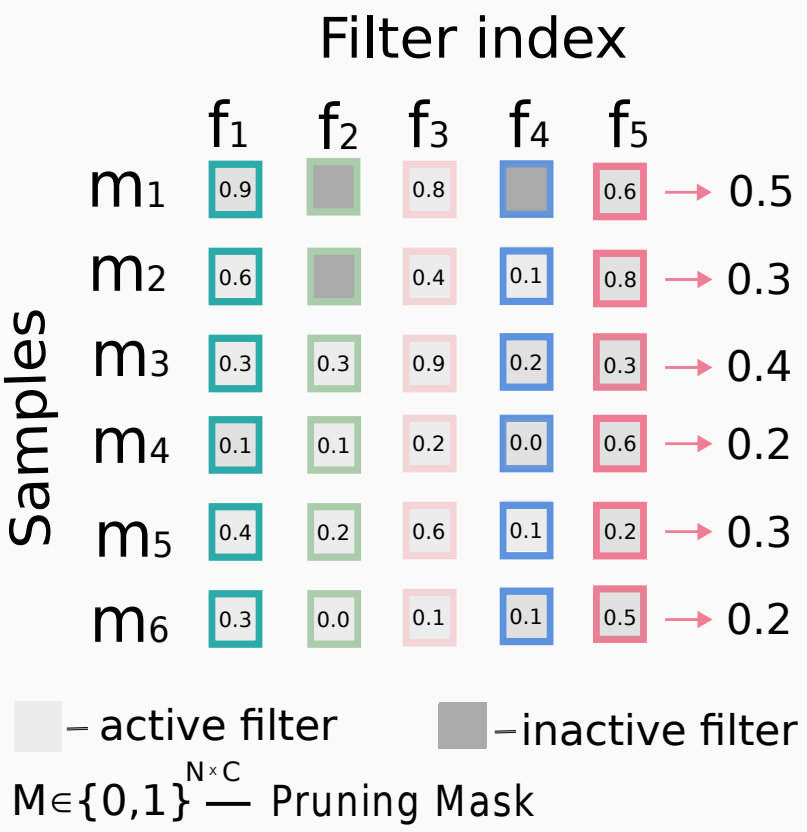

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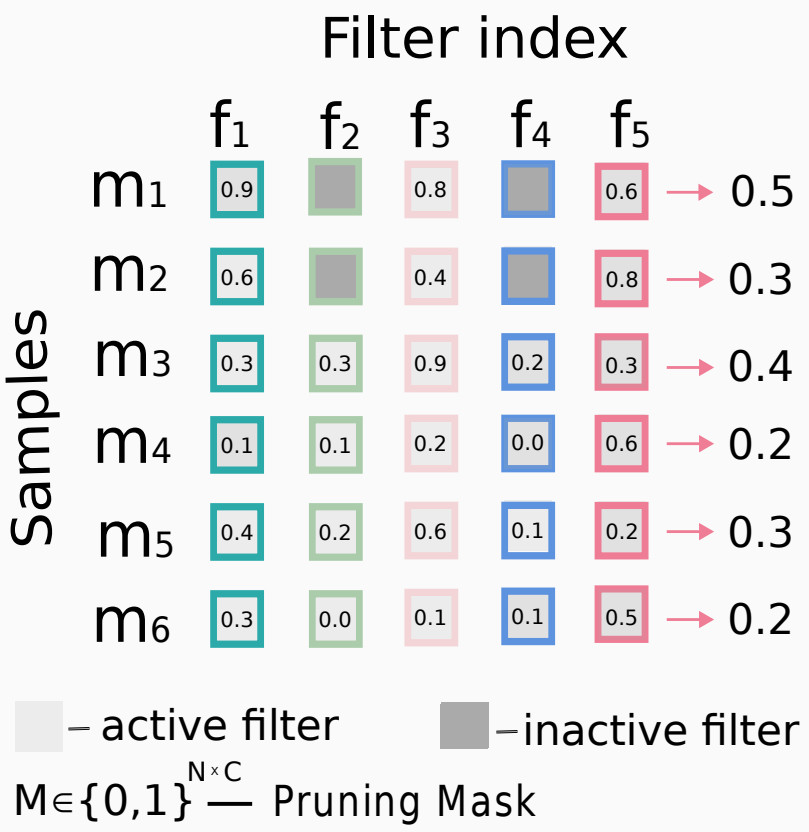

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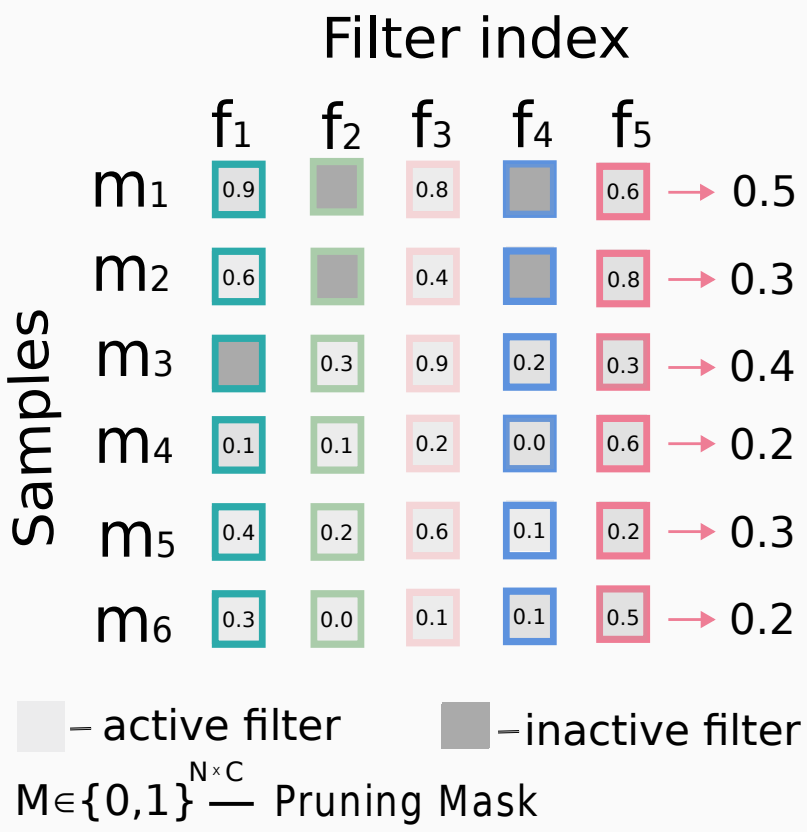

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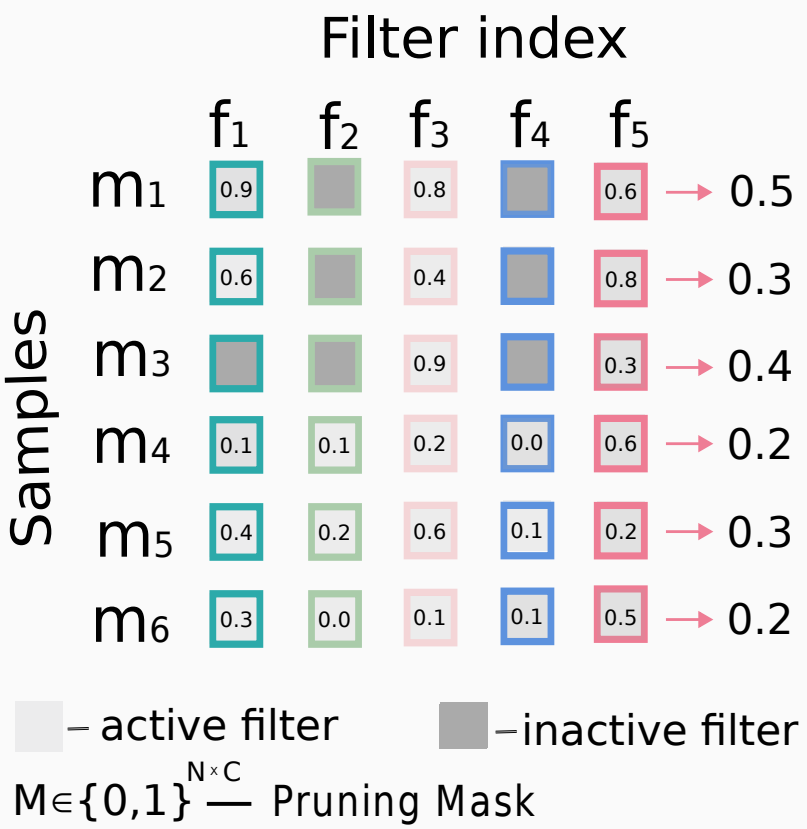

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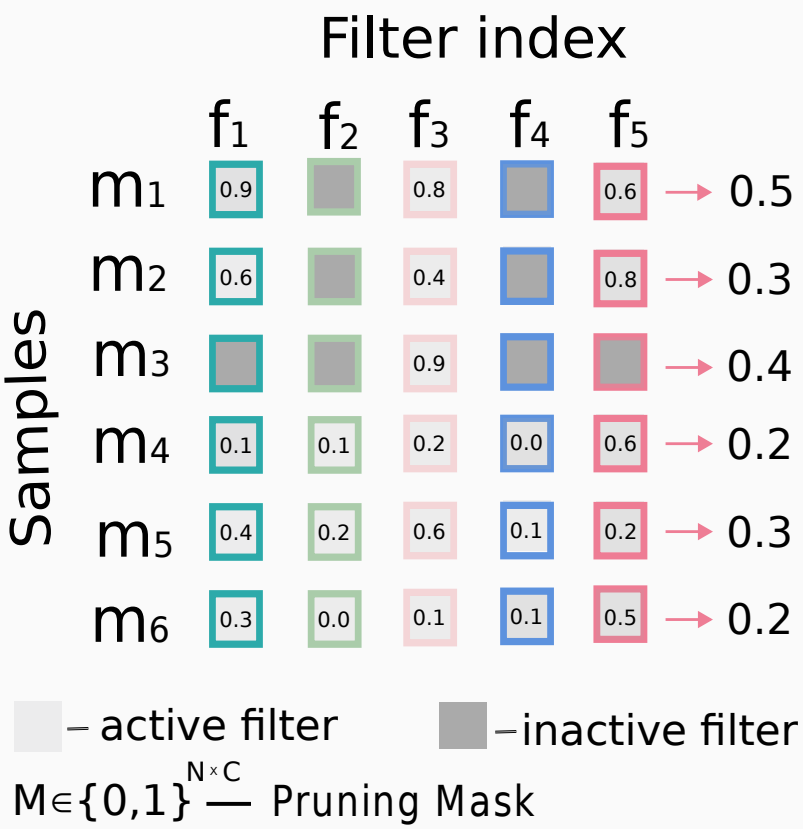

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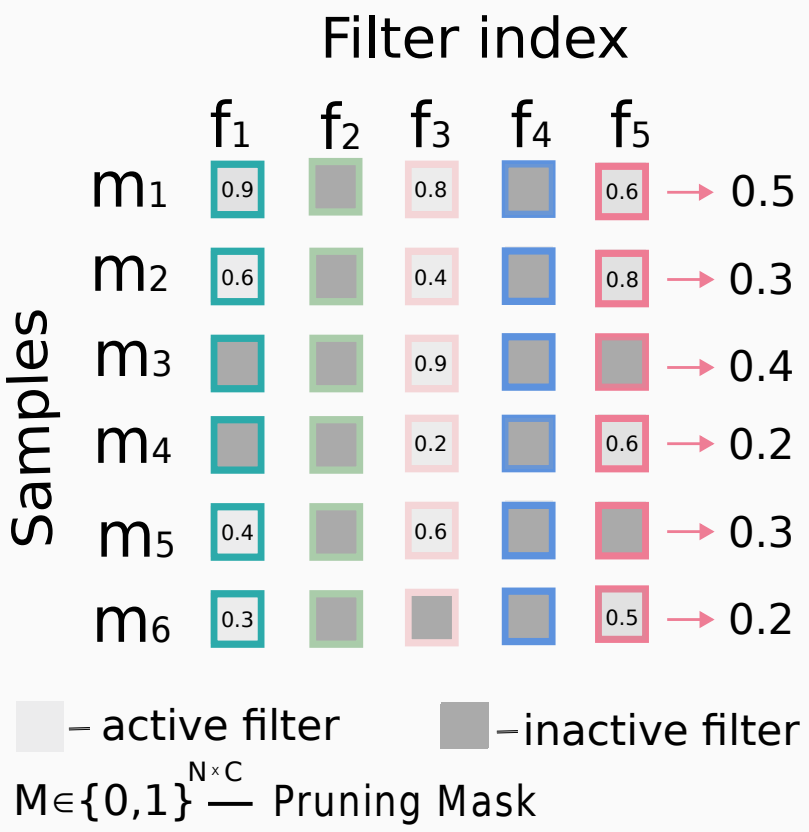

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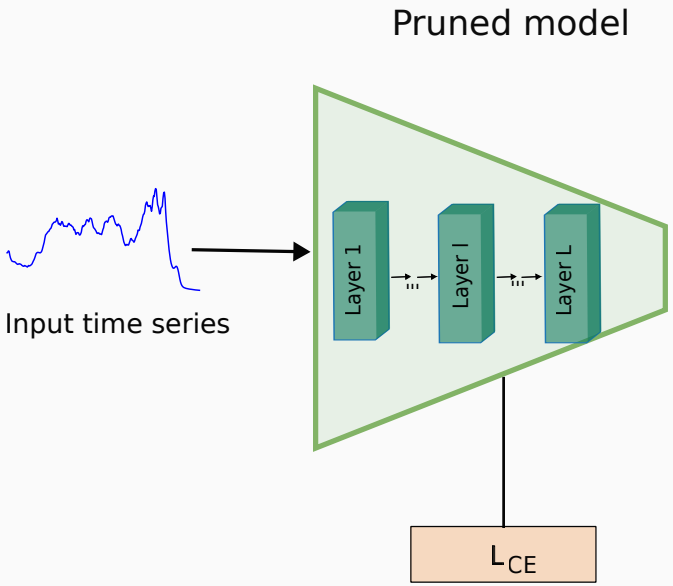
Stage 3: Retraining

Two Retraining Strategies:

- ▷ **Fine-tuning:** Update from pretrained weights
- ▷ **Scratch training:** Reinitialize and retrain



Scratch training recovers performance after aggressive pruning

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Experimental Setup

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  - 5 runs per model, ensemble evaluation
  - LITETime and InceptionTime

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  - Classification accuracy (ensemble of 5 runs)
  - Number of wins, ties and losses computed for pairwise comparisons

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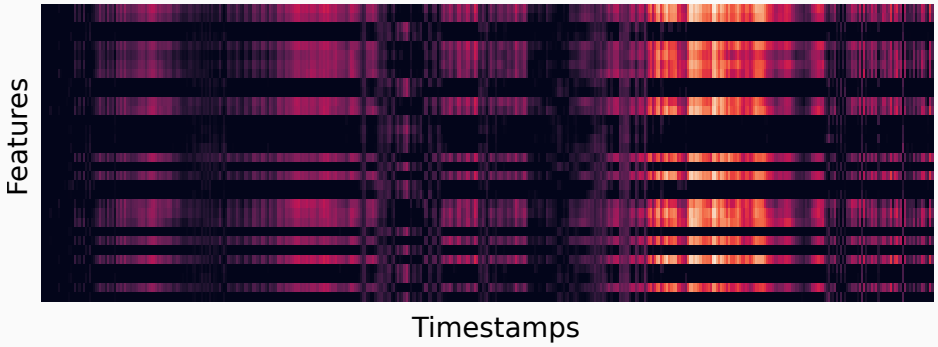
Qualitative Redundancy Analysis

Base LITE Model



- Many active channels
- Redundant patterns

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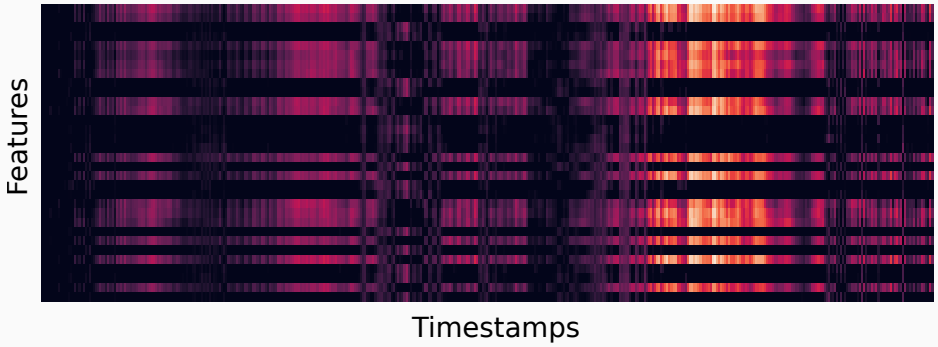

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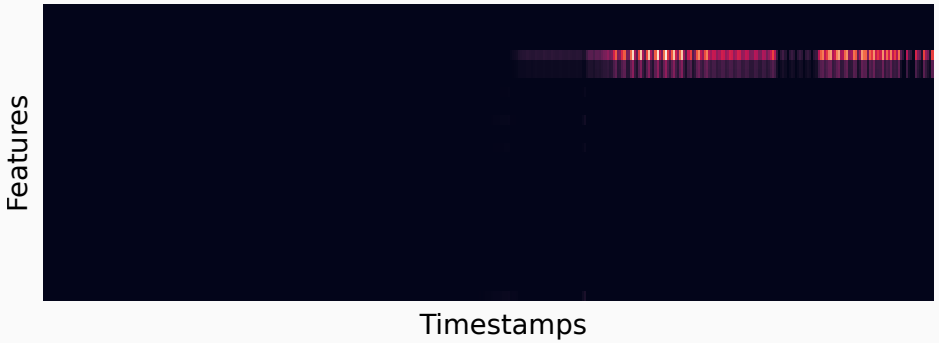
Sparsity-Trained LITE Model



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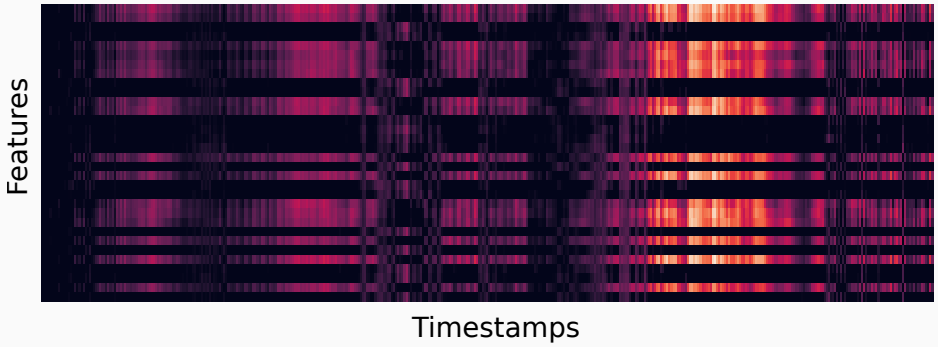

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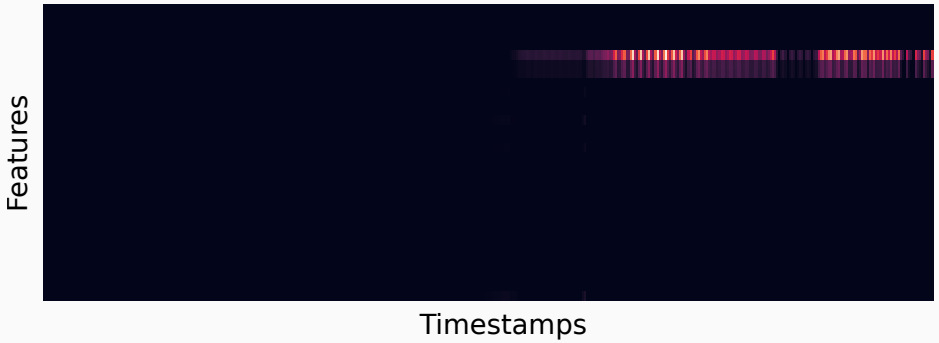
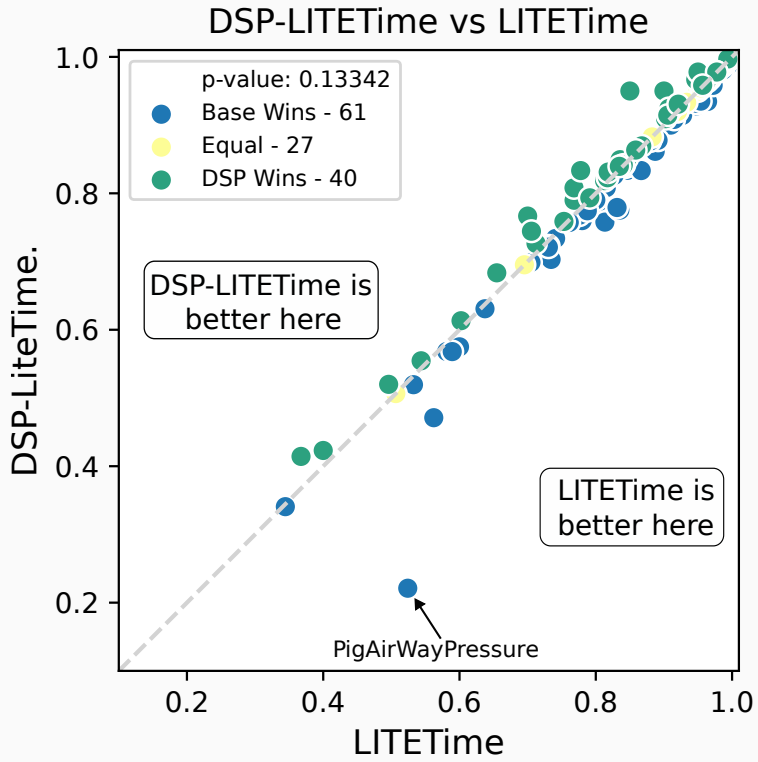


- Compact representation
- Discriminative features

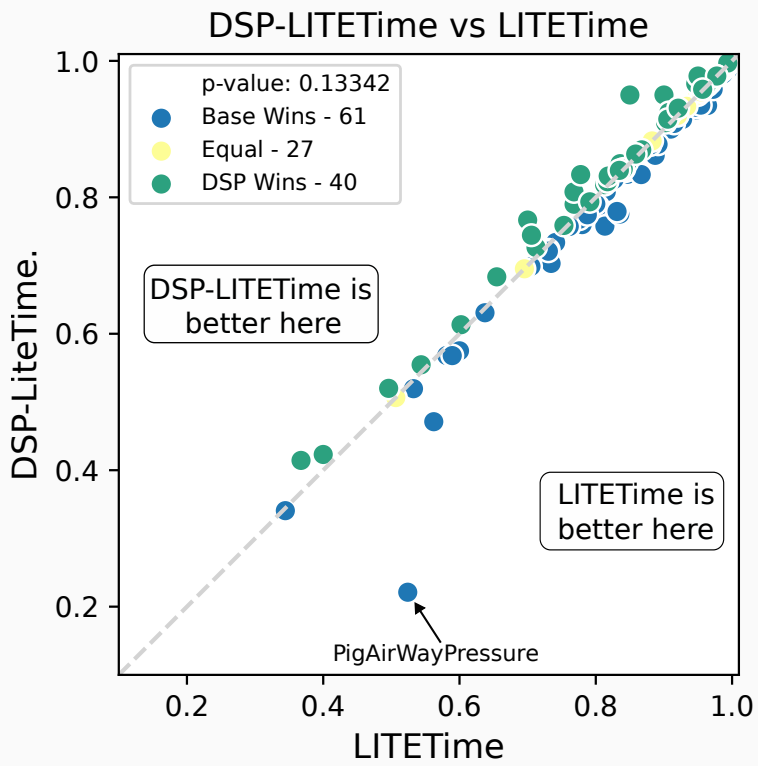
Sparsity loss eliminates redundancy while preserving discriminative patterns

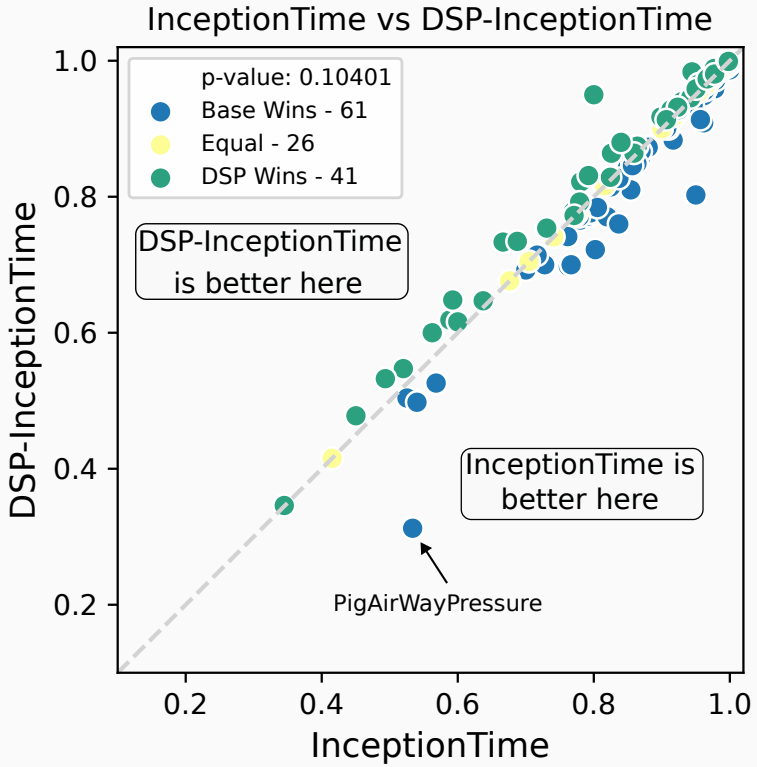
Performance Comparison



- ▷ **58% compression** maintained

Performance Comparison

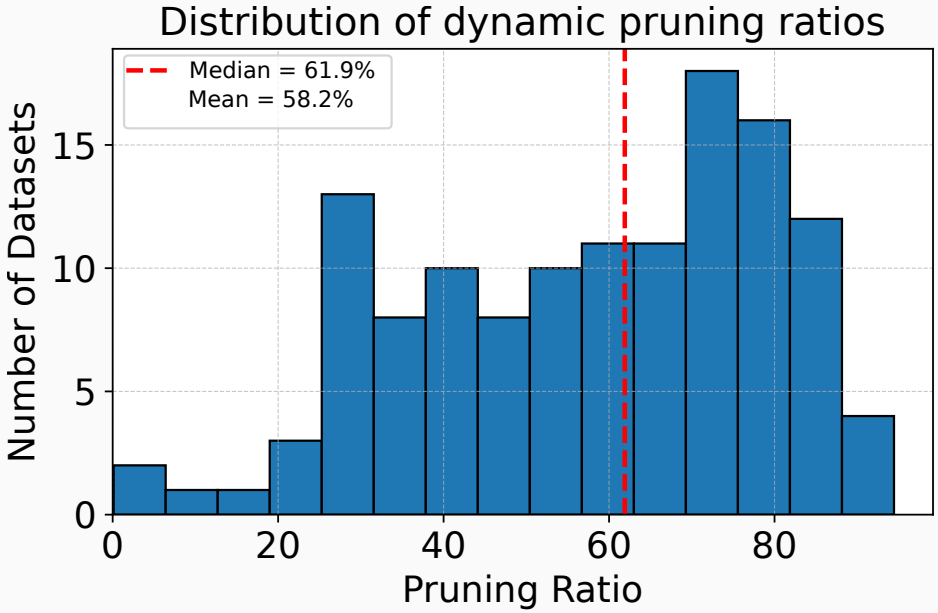




- ▷ **58% compression** maintained
- ▷ **75% compression** maintained

Adaptive Pruning Ratios

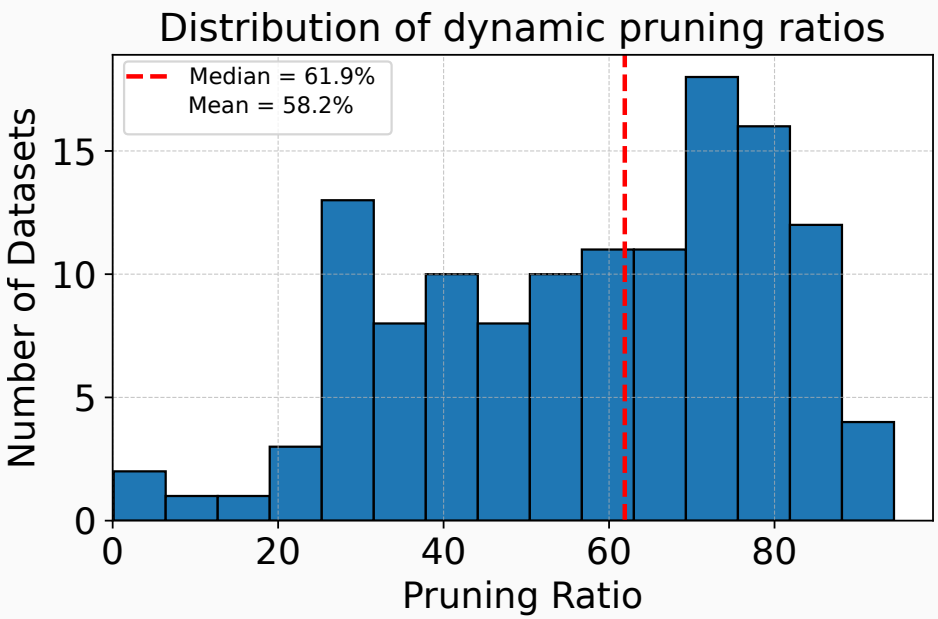
LITETime Model



- Range: 0.01% - 90%

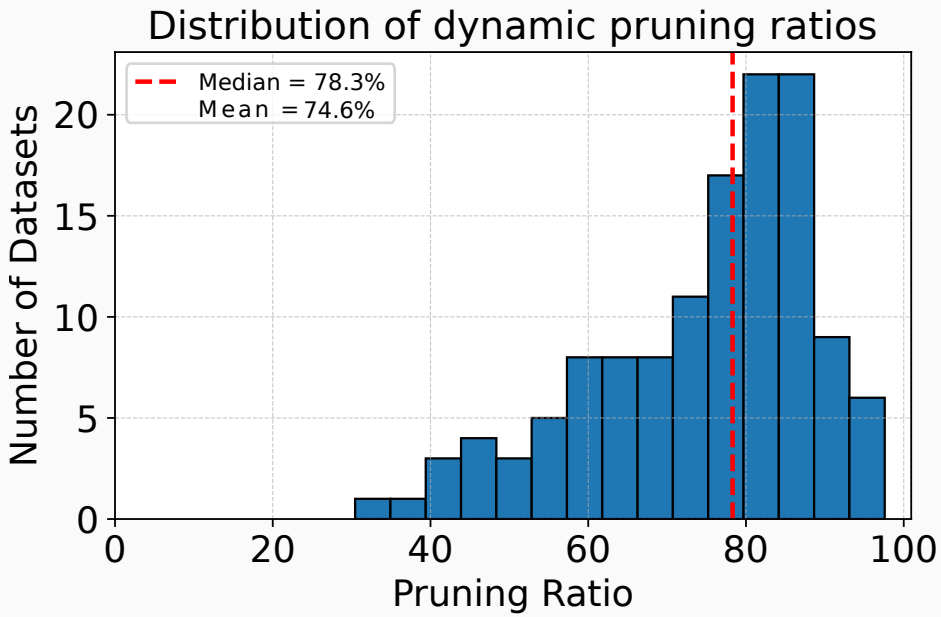
Adaptive Pruning Ratios

LITETime Model



▷ Range: 0.01% - 90%

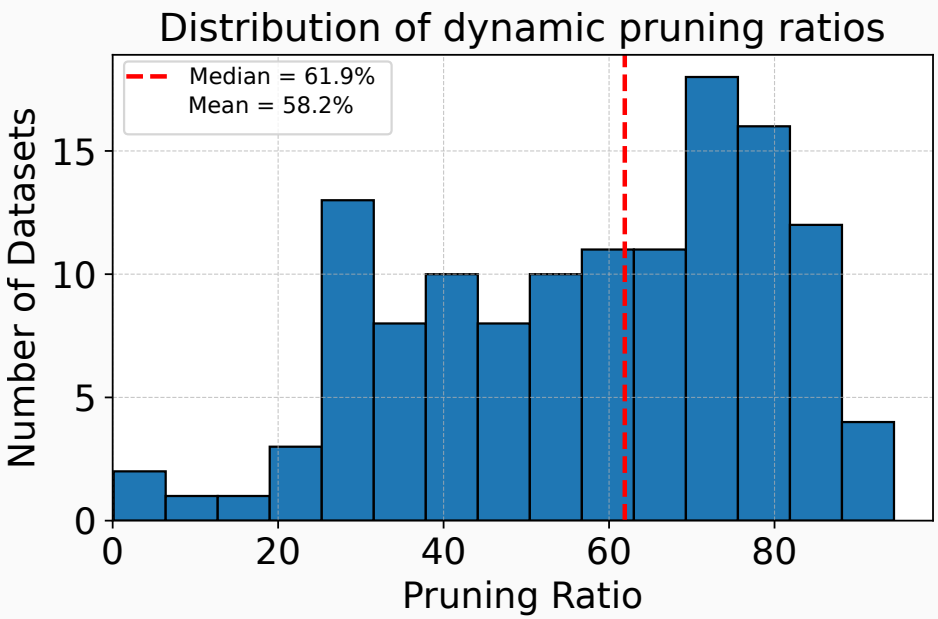
InceptionTime Model



▷ Range: 30.4% - 97.6%

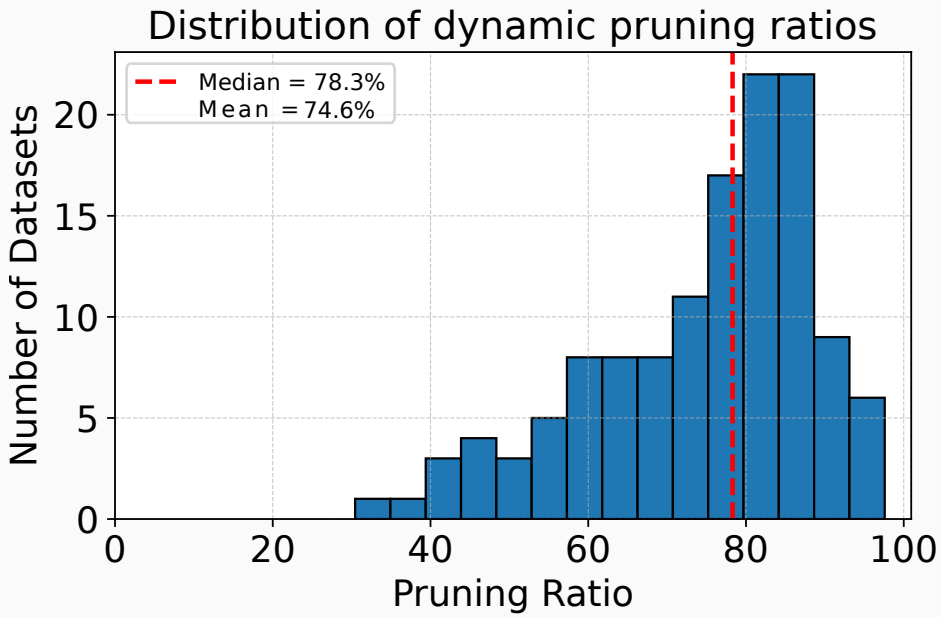
Adaptive Pruning Ratios

LITETime Model



- Range: 0.01% - 90%

InceptionTime Model

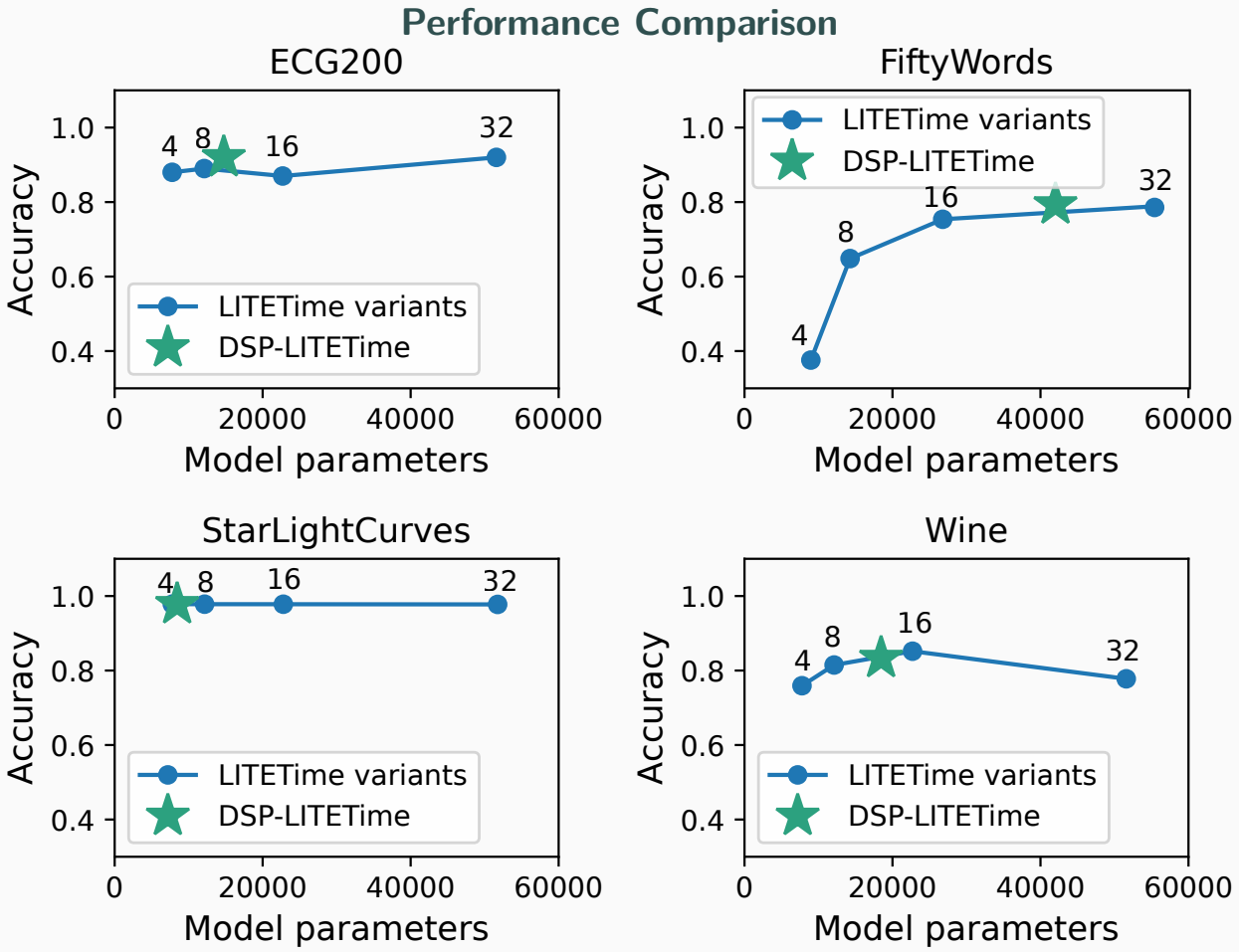


- Range: 30.4% - 97.6%

Key Insights:

- DSP dynamically adapts pruning based on dataset complexity
- Larger model (Inception) → more aggressive pruning
- No manual hyperparameter tuning required

Dynamic vs Static Pruning



Takeaway

- We proposed Dynamic Structured Pruning (DSP), a fully automatic pruning framework for TSC models
- DSP achieves 58% compression for LITETime and 75% for InceptionTime while maintaining accuracy
- No manual hyperparameter tuning required, adapts dynamically to dataset complexity
- Current limitation: validated only on CNN architectures
- Future work: extend to Transformers and combine with quantization and knowledge distillation

Code available here: https://github.com/MSD-IRIMAS/Pruning4TSC

Contact: javidanabdullayev.github.io